

fast buses require high-performance test and measurement systems, which presents a tremendous challenge and business opportunity for test and measurement companies. The typical test needs are for testing the transmitter and the receiver, and also validating the connection medium (cables *et al.*). For transmitter's physical-layer testing, ultra-high-bandwidth oscilloscopes with low noise floor and low intrinsic jitter are required.

Sanchit Bhatia, digital applications specialist, Agilent Technologies, cites an example: "Fibre Channel 28G requires a 45GHz real-time scope for electrical validation and SAS-3 requires a 20GHz oscilloscope. Moving on to the receiver validation, bit-error-rate testers operating at up to 28G data rate are required. These testers stress the receiver by applying calculated jitter and measure the bit error rate. Protocol validation is also done in these high-speed buses through custom-designed high-performance protocol analysers."

Big Data opens three interesting areas for test and measurement: "First, test and measurement has so far been limited to labs and manufacturing lines. With Big Data catching fire, the boundaries of time and distance have broken down and test and measurement now happens on the field. Some applications that have caught on very significantly are online condition monitoring of systems (in other words, aggregation of data). Second, remote monitoring, testing and diagnostics of systems deployed in remote locations without physical access have gained a lot of leverage, which means easier access to data. Last, near-infinite com-

puting resources in the cloud provide an opportunity for software to offload computationally heavy tasks. These can be sophisticated image or signal processing or even compilation and development, which, in short, is 'off-loading,' explains Satish Mohanram, technical marketing manager, National Instruments India.

"Differentiation is no longer about who can collect the most data; it's about who can quickly make sense of the data they collect" – Measurements Outlook 2013, National Instruments Corporation.

Challenges encountered with the 'Big' in Big Data

There used to be a time when hardware sampling rates were limited by the speed at which analogue-to-digital conversion took place, physically restricting the amount of data that could be acquired. But today, hardware is no longer a limiting factor in acquisition applications. Management of the acquired data is the challenge of the future.

As it becomes both necessary and easier to capture large amounts of data at high speeds, engineers will, needless to say, face challenges in creating end-to-end solutions that require a close relationship between the automated test and IT equipment. This is driving test and measurement system providers to work with IT providers in order to offer bundled and integrated solutions for automated test applications.

Contextual data mining. Data mining is the practice of using the contextual information saved along with data to search through and pare down large data sets into more manageable, applicable volumes. By storing raw data alongside its original context, or 'meta-data,' it becomes easier to accumulate, locate and later on, manipulate and comprehend. This is one of the major benefits that makes it easier to analyse the collected data.

Intelligent data acquisition nodes. Though it is common to stream test data to a host PC over standard buses like Ethernet and USB, high-channel-count measurements with fast sampling rates can easily overload the



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